Sang Hak Lee

Marine Data Engineer · Vessel & weather data collection, datasets and quality

vin9604@gmail.com
fortranmentis.github.io
github.com/fortranmentis
linkedin.com/in/sang-hak-lee-ba11a5447

Data engineer who **collects and cleans** AIS / V-PASS vessel data and marine-weather forecast data (NetCDF), and **builds datasets** on PostgreSQL data models I design. Embeds spec-based **data quality validation** and processing **metadata management** into pipelines; licensed weather forecaster with meteorology and ocean-environment certifications, bringing domain understanding of vessel, weather and route data.

EXPERIENCE

GeoSR (Geo System Research) · Data Engineer

Feb 2023 - Present

- **Vessel data collection & cleansing** — Cleaned TB-scale AIS trajectories for the Busan area (outliers, duplicates, time alignment) and joined route/anchorage data to build anchoring, loitering and route-deviation candidate datasets (GeoPandas, multiprocessing).
- **V-PASS dataset construction** — Streaming merge and filtering of vessel observations (data years 2023-2025) into regional/grid statistical datasets and density maps, joined with catch-grade data.
- **Weather & ocean data pipeline** — Automated NetCDF/CSV ingestion → spec validation → raster and current-vector conversion → PostgreSQL loading → GeoServer publishing (Python, Shell, Docker, air-gapped).
- **Data modeling** — Split loading for 11 monitoring/forecast products from 4 institutions into a **file catalog** (data type, year-month, monitoring/forecast, publication flag), **validation results** (per-file pass/fail and error items) and **value tables** (current lat/lon, speed, direction, depth), linked by file name.
- **Data quality validation** — Automated spec checks (Class, Variable, Level, Unit) with error records, exclusion and count reconciliation, service-point to model-grid matching, old vs. new product comparisons.
- **Metadata management** — Maintained processing/error history, input spec sheets and data specifications; traced source file → validation result → loaded values per file.
- **Systems & documentation** — Built forecast production servers (Oracle DB, Tomcat), scheduling, monitoring, incident response; requirements and test documents; conference poster (2024), workshop talk (2025).

Korea Meteorological Industry Promotion Agency · R&D Planning Intern

Jun - Sep 2022

- Researched weather technology and policy trends; wrote monthly newsletters and quarterly reports.

SELECTED PROJECTS

Marine Travel AI Agent · GeoSR internal R&D

2026

Managed public data consumed by an LLM agent — automated TourAPI and 8 marine-forecast feed collection (n8n), duplicate-ID, coordinate-range, missing-page and per-feed freshness checks, collection/issue timestamps, typed validation tools for LLM use (LangGraph, FastAPI).

AIS Trajectory Analytics

2024 - 2025

Activity-candidate detection from AIS tracks with route and anchorage polygons; KOSMEE autumn conference poster (2024) and workshop talk (2025).

Ocean Forecast GeoPipeline

2025.06 - 2026.04

NetCDF/CSV spec validation → spatial conversion → PostgreSQL loading → GeoServer publishing; code and validation docs public on GitHub.

SKILLS

Data	Python, SQL, PostgreSQL, Oracle, Pandas, GeoPandas
Formats	NetCDF, AIS, V-PASS, GeoTIFF, GeoServer, QGIS
Infra	Linux, Shell, Docker, scheduling & monitoring
Other	Fortran, Rust, LangGraph, FastAPI, Ollama (LLM agents)

CREDENTIALS · EDUCATION

Weather Forecaster License · Engineer Meteorology · Engineer Ocean Environment

B.S. Earth & Environmental Sciences, Jeonbuk National University (2016 - 2022)

Research Club Grand Prize, JBNU College of Natural Sciences (2021)